

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

$$\begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$$

=

$$\begin{pmatrix} 26 \\ 62 \\ 98 \end{pmatrix}$$

$$\begin{array}{rcccc} j: & 1 & 2 & 3 & \\ & 1 & 2 & 3 & a_{1j} \\ & | & | & | & \\ & 3 & 4 & 5 & b_j \end{array}$$

$$3 + 8 + 15 = 26$$